

Class17

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Section 1. Proportion of G/G in a population

Here we start by reading the downloaded CSV file.

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

	Sample..Male.Female.Unknown.	Genotype..forward.strand.	Population.s.	Father
1		NA19648 (F)	A A ALL, AMR, MXL	-
2		NA19649 (M)	G G ALL, AMR, MXL	-
3		NA19651 (F)	A A ALL, AMR, MXL	-
4		NA19652 (M)	G G ALL, AMR, MXL	-
5		NA19654 (F)	G G ALL, AMR, MXL	-
6		NA19655 (M)	A G ALL, AMR, MXL	-
	Mother			
1	-			
2	-			
3	-			
4	-			
5	-			
6	-			

```
table(mxl$Genotype..forward.strand.)
```

```
A|A A|G G|A G|G
22 21 12 9
```

```
table(mx1$Genotype..forward.strand.)/nrow(mx1)
```

```
      A|A      A|G      G|A      G|G
0.343750 0.328125 0.187500 0.140625
```

Now, let's look at a different population, GBR.

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(gbr)
```

	Sample..Male.Female.Unknown.	Genotype..forward.strand.	Population.s.	Father
1		HG00096 (M)	A A ALL, EUR, GBR	-
2		HG00097 (F)	G A ALL, EUR, GBR	-
3		HG00099 (F)	G G ALL, EUR, GBR	-
4		HG00100 (F)	A A ALL, EUR, GBR	-
5		HG00101 (M)	A A ALL, EUR, GBR	-
6		HG00102 (F)	A A ALL, EUR, GBR	-
	Mother			
1	-			
2	-			
3	-			
4	-			
5	-			
6	-			

```
table(gbr$Genotype..forward.strand.)/nrow(gbr)
```

```
      A|A      A|G      G|A      G|G
0.2527473 0.1868132 0.2637363 0.2967033
```

This variant that is associated with childhood asthma is more frequent in the GBR population than the MKL population.

Lets now dig into this further.

Section 2. Population scale analysis (Homework)

Now, let's start some population scale analysis

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
table(expr$geno)
```

A/A	A/G	G/G
108	233	121

Therefore, we have 108 A/A individuals, 233 A/G individuals and 121 G/G individuals.

```
tapply(expr$exp, expr$geno, median)
```

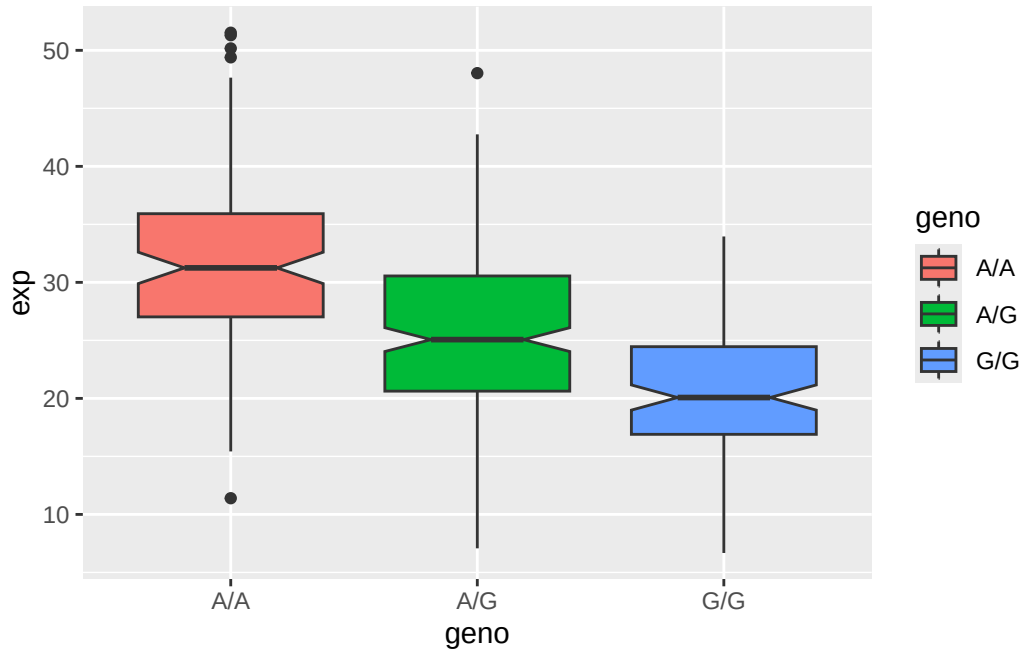
A/A	A/G	G/G
31.24847	25.06486	20.07363

Therefore, the median expression level for A/A is 31.24847, for A/G is 25.06486 and for G/G is 20.07363.

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
library(ggplot2)

ggplot(expr) + aes(geno, exp, fill = geno) +
  geom_boxplot(notch = TRUE)
```



The boxplot shows a clear, graded difference in ORMDL3 expression across the three genotypes. Individuals with the A/A genotype have the highest median expression, A/G individuals show an intermediate median and G/G individuals have the lowest median expression. The notches in the boxplot do not overlap substantially between A/A and G/G, indicating that their medians are significantly different.

Therefore, the SNP appears to have a strong effect on ORMDL3 expression: carrying the G allele is associated with decreased ORMDL3 expression, while the A allele is associated with higher expression.